

QUESTION 2

A famous ski-resort has been built in "Avalanche Valley". This resort is open to skiers on Saturday and Sundays, but is closed to skiers Monday through Friday. Over the past 10 years, the owners have recorded the number of avalanches reported on each day and have seen the following numbers. **NOTE** - You can assume that these numbers are based on the same number of days i.e. Exactly 520 Mondays, 520 Tuesdays, ..., 520 Sundays over the past 10 years

On Mondays - 10 avalanches

On Tuesdays - 12 avalanches

Wednesday - 15 avalanches

Thursday - 16 avalanches

Friday - 8 avalanches

Saturday - 22 avalanches

Sunday - 21 avalanches

The Environmental Protection Agency is worried that avalanches are more likely to occur on days when people are skiing.

A. Given that 104 avalanches occurred during the past 10 years:

- i) what is the expected number of avalanches that should happen on any given day of the week if there is no association between day and probability of avalanche and
- ii) give a large sample 95% confidence interval for the number of avalanches seen on a given day of the week if there is no association between day and probability of avalanche

B. Based on 104 avalanches occurring overall, perform the appropriate two-sided large sample test of H_0 : that the probability of an avalanches is independent of whether or not people are skiing on that day (remember that people are skiing on Saturdays & Sundays only) at $\alpha = 0.01$